



Question Bank

NET Centric Question 2078



Tribhuvan University Institute of Science and Technology 2078

Bachelor Level / Science

Full Marks: 60 + 20 + 20

Bsc. CSIT

Pass Marks: 24 + 8 + 8

NET Centric Computing

Time: 3 Hours

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

Attempt any two questions.

1. Differentiate between struct and enum. Why do we need to handle the exception? Illustrate with an example with your own customized exception. ✓

2. Explain the procedure for building, running, and deploying .NET core applications. Write a program to show ✓

Differentiate between struct and enum. Why do we need to handle the exception? Illustrate with an example with your own customized exception.

Expert approved answer

Struct	Enum
The “struct” keyword is used to declare a structure.	The “enum” keyword is used to declare an enumerated datatype.
A struct can contain both data variables and methods.	Enum can only contain data types.
A struct supports a private but not protected access specifier.	Enum doesn’t have private and protected access specifiers.
Struct supports encapsulation.	Enum doesn’t support encapsulation.
When the	Once the enum is



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When the structure is declared, the values of its objects can be modified.	Once the enum is declared, its value cannot be changed, otherwise, the compiler will throw an error.
Struct only contains parameterized constructors and no destructors.	Enum doesn't contain constructors and destructors.
Value Type.	Reference Type.

Exception handling is an essential aspect of programming in C# because it allows you to easily handle unexpected or exceptional situations that can occur during the execution of a program. By handling exceptions, you can prevent your application from crashing or displaying cryptic error messages to the user. Instead, you can provide meaningful feedback and take appropriate actions to recover from the



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Console class Program



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Instead, you can provide meaningful feedback and take appropriate actions to recover from the exceptional condition.

Console. class Program

```
{
static void Main()
{
try
{
int dividend = 10;
int divisor = 0;
int result = dividend / divisor;
Console.WriteLine("Result: " + result);
}
catch (DivideByZeroException ex)
{
WriteLine("Error: Cannot divide by zero.");
}
}
}
```

In this example, we have a simple division

or

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set the divisor to zero, which will result in a





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```
catch (DivideByZeroException ex)
{
    WriteLine("Error: Cannot divide by zero.");
}
}
```

In this example, we have a simple division operation where we attempt to divide a dividend by a divisor. However, we deliberately set the divisor to zero, which will result in a division by zero error. To handle this exceptional situation, we wrap the division operation with a try-catch block. Inside the catch block, we catch the `DivideByZeroException`, which is a built-in exception specifically designed to handle division by zero errors. If this exception occurs, we display a custom error message stating that division by zero is not allowed.



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Explain the procedure for building, running, and deploying .NET core applications. Write a program to show the possibility of SQL injection attack.

Expert approved answer

Procedure for Building, Running, and Deploying .NET Core Applications:

1. Building a .NET Core Application

- Create a new .NET Core project using the ``dotnet new`` command or your preferred IDE.
- Write the code for your application.
- Use the ``dotnet build`` command to build the project. This command compiles the code and generates the executable or assembly.

2. Running a .NET Core Application

- Use the ``dotnet run`` command to run the application. This compiles and executes the code.



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code.

– Alternatively, you can navigate to the output directory and run the generated executable directly.

3. Deploying a .NET Core Application

- Choose a deployment target based on your application's requirements, such as a web server
- Publish your application using the ``dotnet publish`` command. This command packages the application and its dependencies into a self-contained directory.
- Transfer the published files to the deployment target.
- Configure the deployment target to run the application.

SQL Injection Attack Example:

SQL injection is a security vulnerability that occurs when an attacker can manipulate user-

SQL injection is a security vulnerability that occurs when an attacker can manipulate user-

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the possibility of a SQL injection attack, here's



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occurs when an attacker can manipulate user-supplied input to execute arbitrary SQL commands on a database. To demonstrate the possibility of a SQL injection attack, here's an example:

```
using System;
using System.Data.SqlClient;
class Program
{
    static void Main()
    {
        Console.Write("Enter a username: ");
        string username = Console.ReadLine();
        string connectionString =
            "your_connection_string";
        string query = $"SELECT * FROM Users
            WHERE Username = '{username}'";
        using (SqlConnection connection = new
            SqlConnection(connectionString))
        {
            connection.Open();
            SqlCommand command = new
```

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```
SqlCommand command = new
SqlCommand(query, connection);
SqlDataReader reader =
command.ExecuteReader();
while (reader.Read())
{
    string userId = reader["UserId"].ToString();
    Console.WriteLine("User ID: " + userId);
}
}
```

In this example, we have a program that asks the user to enter a username. The program constructs a SQL query to retrieve the user's ID from a database table called "Users" based on the provided username. However, this code is vulnerable to a SQL injection attack. An attacker can input malicious characters or SQL statements that modify the intended behaviour of the query. For example, an attacker can enter ' OR 1=1 -- ' as the

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SELECT * FROM Users WHERE Username =



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is vulnerable to a SQL injection attack. An attacker can input malicious characters or SQL statements that modify the intended behaviour of the query. For example, an attacker can enter ' OR 1=1 – ' as the username, which would result in a query like:

```
SELECT * FROM Users WHERE Username = '
OR 1=1 –'
```

In this case, the attacker's intention is to retrieve all user records from the table by exploiting the 'OR 1=1' condition, which is always true, and the – comments out the rest of the query. To prevent SQL injection attacks, it is crucial to use parameterized queries or stored procedures. Parameterized queries allow you to separate the SQL code from the user input, ensuring that the input is treated as data rather than executable code.



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What are the advantages of stateless HTTP in ASP.NET core application? Create a method to insert a record (3, "John , 12000") onto the table Employee having fields Employeeid(int), Name varchar(200), Salary(int) using Entity Framework.

Expert approved answer

Advantages of Stateless HTTP in ASP.NET Core Applications:

- Stateless HTTP allows for easy scalability since each request can be processed independently without relying on the state stored on the server. – With stateless HTTP, the server doesn't need to manage and track session state for each client. This reduces the overhead associated with session management
- Statelessness simplifies the application architecture and reduces the chances of bugs



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- Statelessness simplifies the application architecture and reduces the chances of bugs related to shared state or session management. This improves the overall reliability and stability of the application.
- Stateless HTTP is more cache-friendly since responses can be easily cached and served to multiple clients. This reduces the load on the server and improves the efficiency of the application.
- Stateless HTTP makes it easier to test individual requests and responses since each request is self-contained and doesn't rely on any shared state.

using System;

using Microsoft.EntityFrameworkCore;

// Define the Employee model class

class Employee

{



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```
class Employee
```

```
{
```

```
public int EmployeeId { get; set; }
```

```
public string Name { get; set; }
```

```
public int Salary { get; set; }
```

```
}
```

```
class Program
```

```
{
```

```
static void Main()
```

```
{
```

```
// Create a new instance of DbContextOptions  
with the connection string
```

```
var options = new DbContextOptionsBuilder()
```

```
.UseSqlServer(connectionString);
```

```
Options;
```



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```
.UseSqlServer("your_connection_string")
```

```
.Options;
```

```
// Create a new instance of  
EmployeeDbContext with the options
```

```
using (var dbContext = new  
EmployeeDbContext(options))
```

```
{
```

```
// Create a new Employee record
```

```
var newEmployee = new Employee
```

```
{
```

```
EmployeeId = 3,
```

```
Name = "John",
```

```
Salary = 12000
```

```
};
```



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```
//
```





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```
Name = Jonn ,
```

```
Salary = 12000
```

```
};
```

```
// Add the new Employee record to the context
```

```
dbContext.Add(newEmployee);
```

```
// Save the changes to the database
```

```
dbContext.SaveChanges();
```

```
Console.WriteLine("Record inserted  
successfully.");
```

```
}
```

```
}
```

```
}
```



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Best Answer of the Question

Define open redirect attacks. Discuss about docker and containerization.

Expert approved answer

Open redirect attacks are a type of security vulnerability that occurs when a web application allows an attacker to redirect users to an arbitrary URL. This can be exploited by an attacker to trick users into visiting malicious websites or phishing pages, potentially leading to various types of attacks, such as stealing sensitive information, distributing malware, or gaining unauthorized access to user accounts. The vulnerability typically arises when a web application includes a parameter in the URL that determines the destination of a redirect. If this parameter can be manipulated by an attacker, they can change the redirect URL to an external site under their control.

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Docker and Containerization:

Docker is an open-source platform that enables developers to automate the deployment and management of applications within lightweight, isolated containers.

Containerization is a technology that allows you to package an application along with its dependencies and configuration files into a standardized unit called a container.

Containers provide a consistent and portable environment for running applications across different computing environments, such as development, testing, and production.

Key aspects and benefits of Docker and containerization include:

- Containers provide process-level isolation, allowing applications to run independently of the host system and other containers.
- Containers are highly portable, as they package applications with their dependencies and configuration. This makes it easier to



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environments, such as from a developer's



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different computing environments, such as development, testing, and production.

Key aspects and benefits of Docker and containerization include:

- Containers provide process-level isolation, allowing applications to run independently of the host system and other containers.
- Containers are highly portable, as they package applications with their dependencies and configuration. This makes it easier to move applications between different environments, such as from a developer's machine to a production server, without worrying about compatibility issues.
- Docker enables efficient scaling of applications by allowing you to run multiple containers across different hosts.



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What is named and positional attribute parameters? Describe the .Net architecture design and principles.

Expert approved answer

Named and positional attribute parameters refer to different ways of passing arguments to a function or method.

1. Positional Attribute Parameters: are defined based on their position or order in the function/method signature. When calling a function with positional attribute parameters, the arguments are passed in the same order as defined in the function signature. The values are matched to the corresponding parameters based on their position.

2. Named Attribute Parameters: are passed based on their parameter names rather than their positions. When calling a function with



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2. Named Attribute Parameters: are passed based on their parameter names rather than their positions. When calling a function with named attribute parameters, the arguments are specified with their corresponding parameter names, followed by an equals sign (=) and the argument value.

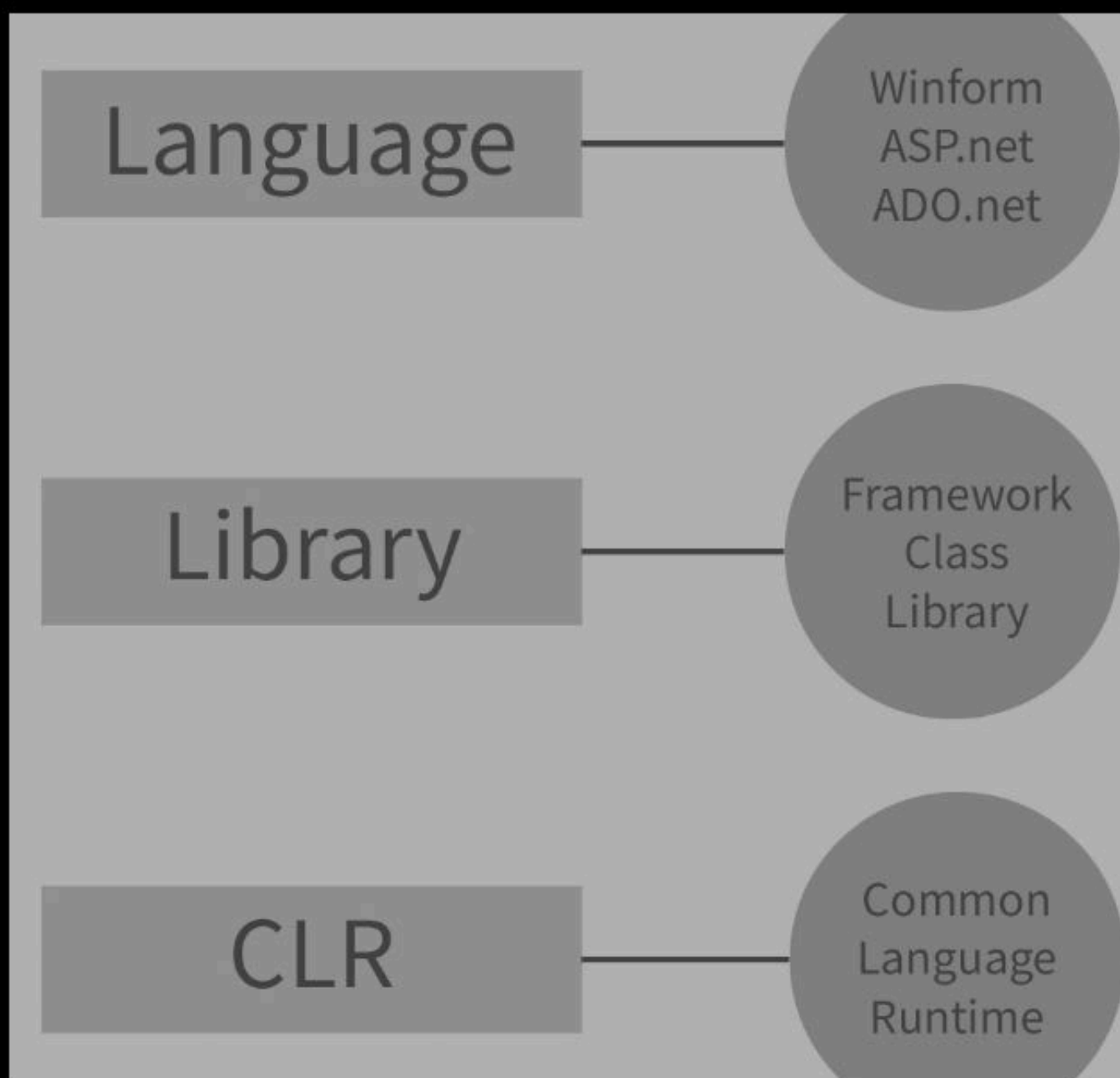


Figure: Net Framework Architecture Diagram

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The following design principles of the .Net



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Net Framework Design Principle

The following design principles of the .Net framework are what make it very relevant to create .Net-based applications.

-Interoperability: The .Net framework provides a lot of backward support. Suppose you had an application built on an older version of the .Net framework, say 2.0. And if you tried to run the same application on a machine that had the higher version of the .Net framework, say 3.5. The application would still work. This is because, with every release, Microsoft ensures that older framework versions will get well with the latest version.

-Portability: Applications built on the .Net framework can be made to work on any Windows platform. And now in recent times, Microsoft is also envisioning making Microsoft products work on other platforms, such as iOS and Linux.



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See also





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and Linux.

-Security: The .Net framework has a good security mechanism. The inbuilt security mechanism helps in both validation and verification of applications. Every application can explicitly define its security mechanism. Each security mechanism is used to grant the user access to the code or to the running program.

-Memory Management: The common language runtime does all work or memory management. The .Net framework has all the capability to see those resources, which are not used by a running program. It would then release those resources accordingly. This is done via a program called the "Garbage Collector" which runs as part of the .Net framework. The garbage collector runs at regular intervals and keeps on checking which system resources are not utilized, and frees them accordingly.



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Explain about request and response message format with example.

Expert approved answer

⊕ HTTP: Request and Response Message Format :

To communicate with a web server, the client, makes calls over the network using HTTP. A client makes an HTTP request for a resource, and the server sends back an HTTP response.

1. User requests a web page by a URL.

`http://notejunction.com/page.html`

5. Browser renders HTML on page.

`http://notejunction.com/page.html`

Welcome to Website!

2. Browser sends HTTP request to server.

HTTP request

4. Server sends HTML in HTTP response back to browser.

HTTP response

```
<HTML>
<HEAD> </HEAD>
<BODY> </BODY>
</HTML>
```

3. Server interprets request and generates appropriate HTML.

→ The user starts by requesting a web page, which causes an HTTP request to be sent to the server. The server interprets the request, generates the necessary HTML, and sends it back in an HTTP response. The browser can then display the web page.



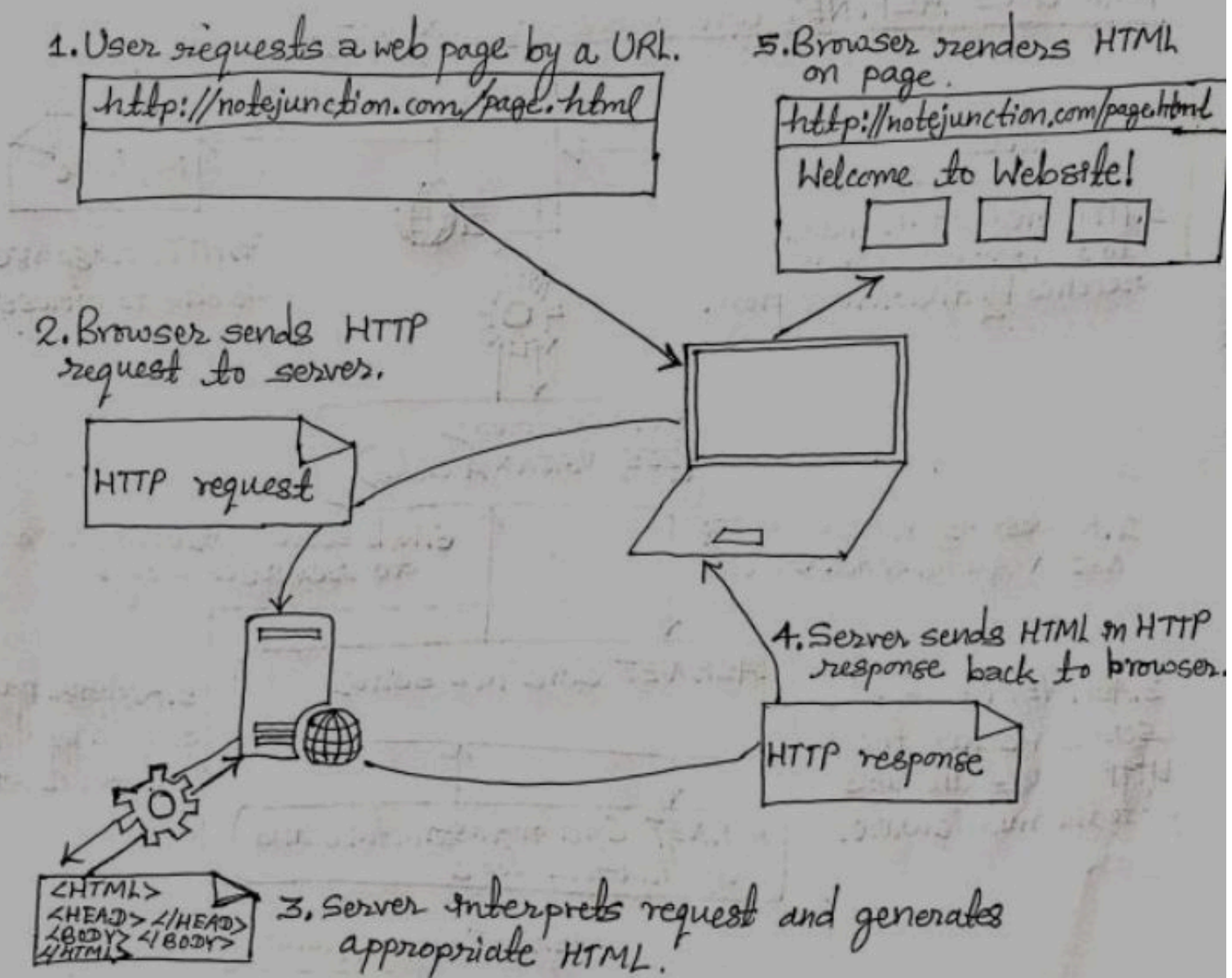
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⊗ HTTP: Request and Response Message Format :

To communicate with a web server, the client, makes calls over the network using HTTP. A client makes an HTTP request for a resource, and the server sends back an HTTP response.



→ The user starts by requesting a web page, which causes an HTTP request to be sent to the server. The server interprets the request, generates the necessary HTML, and sends it back in an HTTP response. The browser can then display the web page.



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Why do we need to validate the form? How can you use react for this?

Expert approved answer

Validating forms in .NET is important to ensure that the data submitted by users is accurate, complete, and meets the expected criteria. Form validation helps prevent data entry errors, improves the user experience by providing timely feedback, and enhances the overall integrity and security of the application. Form validation helps maintain data integrity by ensuring that the data entered by users conforms to the expected format and constraints. It prevents invalid or inconsistent data from being stored in the application's database, avoiding issues with data quality. Proper form validation is also crucial for security purposes. It helps prevent common security vulnerabilities such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). Form validation is a powerful and flexible way to handle form



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common security vulnerabilities such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). React provides a powerful and flexible way to handle form validation through its component-based architecture and state management. React provides various libraries and tools that can assist with form validation, such as Formik, Yup, or React Hook Form. These libraries offer additional features like form state management, validation schema, and error handling helpers, making the form validation process more streamlined and efficient.

An example of name validation in form is given below:

```
import React, { useState } from 'react';  
const FormComponent = () => {  
  const [name, setName] = useState("");  
  const [error, setError] = useState("");  
  const handleSubmit = (e) => {  
    e.preventDefault();  
    if (!name.trim()) {
```

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```
e.preventDefault();
if (!name.trim()) {
  setError('Name is required');
  return;
}
console.log('Form submitted:', { name });
};
return (
  <form onSubmit={handleSubmit}>
    <input
      type="text"
      placeholder="Name"
      value={name}
      onChange={(e) => setName(e.target.value)}
    />
    {error && <div className="error">{error}
    </div>}
    <button type="submit">Submit</button>
  </form>
); };
export default FormComponent;
```



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Define data model and data context. What does URL routing mean?

Expert approved answer

Data Model:

A data model is a conceptual representation of how data is organized and structured within a system or application. It defines the relationships between data entities, their attributes, and the rules governing data manipulation and operations. Data models provide a blueprint for designing databases and serve as a foundation for data management and application development. Common types of data models include relational models, hierarchical models, network models, and object-oriented models.

Data Context:

In the context of software development, a data context refers to an object or component that



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specification, and the context of the data.



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In the context of software development, a data context refers to an object or component that encapsulates the connection and access to a specific data source, such as a database. The data context provides a convenient and centralized way to interact with the underlying data, including querying, inserting, updating, and deleting data. It simplifies data access and manipulation for application developers. In frameworks like Entity Framework in .NET, the data context is often represented by a DbContext class.

URL Routing:

URL routing refers to the process of mapping incoming URL requests to specific endpoints or resources within a web application. It enables the application to handle different URLs and route them to appropriate handlers or controllers, allowing for dynamic content generation and navigation. URL routing is commonly used in web frameworks and platforms to implement features like dynamic



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enables the application to handle different URLs and route them to appropriate handlers or controllers, allowing for dynamic content generation and navigation. URL routing is commonly used in web frameworks and platforms to implement features like dynamic routing, parameterized URLs, and RESTful API routing. It is also used to generate outgoing URLs. This process is handled by the Routing Middleware. Routing Middleware is available in Microsoft.AspNetCore.Routing Namespace



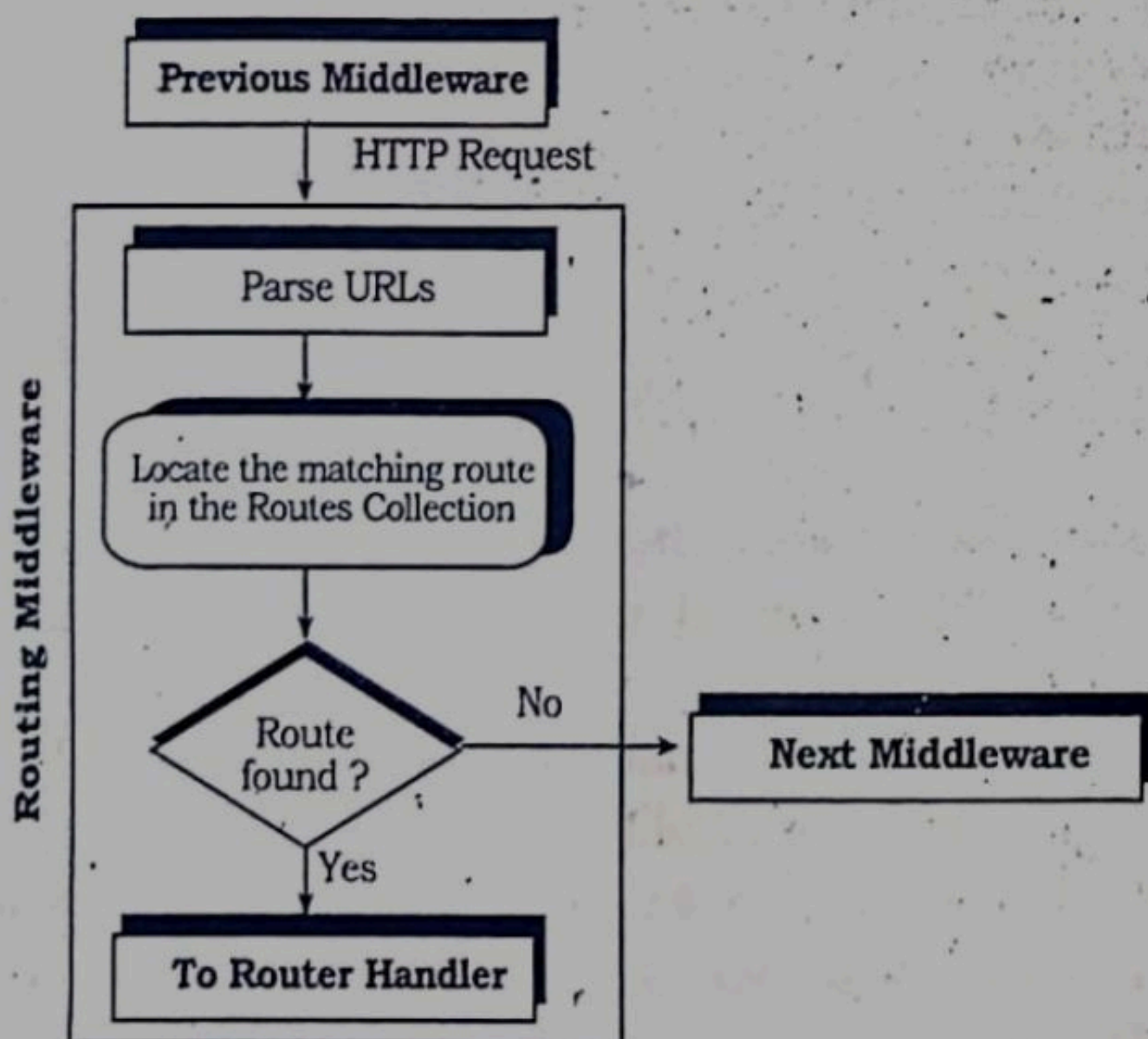
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How Routing Works in ASP.NET Core





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Give an example to manage the session state.

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Session state is an ASP.NET Core mechanism to store user data while the user browses the application. It uses a store maintained by the application to persist data across requests from a client.

Let's create a controller with endpoints to set and read a value from the session:

```
public class WelcomemController : Controller  
  
{  
  
    public IActionResult Index()  
  
    {  
  
        HttpContext.Session.SetString("Name","John");  
  
    }  
}
```

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```
HttpContext.Session.SetString("Name","John");
```

```
HttpContext.Session.SetInt32("Age",32);
```

```
return View();
```

```
}
```

```
public IActionResult Get()
```

```
{
```

```
User newUser = new User()
```

```
{
```

```
Name =
```

```
HttpContext.Session.GetString("Name"),
```

```
Age =
```

```
HttpContext.Session.GetInt32("Age").Value
```

```
};
```

```
return View(newUser);
```

```
}
```



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```
{  
  
    Name =  
    HttpContext.Session.GetString("Name"),  
  
    Age =  
    HttpContext.Sesssion.GetInt32("Age").Value  
  
};  
  
return View(newUser);  
  
}  
  
}
```

The Index() method sets the values into session and Get() method reads the values from the session and passes them into the view.



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Explain the process to deploy the core application.

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Some of the web servers that we can use to deploy our ASP.NET Core Application are:

-Microsoft IIS

-Apache

-NGINX

Steps to deploy ASP.NET Core to IIS

Before we deploy our application, we need to install the .NET Core hosting bundle for IIS – .NET Core runtime, libraries, and the ASP.NET Core module for IIS.

Step 1: Publish to a File Folder.

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Step 1: Publish to a File Folder.

Step 2: Copy Files to the Preferred IIS Location.

For example, let's copy the files to
C:\inetpub\wwwroot\AspNetCore46

Step 3: Create an Application in IIS.

While creating an application in IIS is listed as a single "Step", you will take multiple actions. First, create a new IIS Application Pool under the .NET CLR version of "No Managed Code". Since IIS only works as a reverse proxy, it isn't actually executing any .NET code. Second, you can create your application under an existing or a new IIS Site. Either way, you will want to pick your new IIS Application Pool and point it to the folder you copied your ASP.NET publish output files to.

Step 4: Load the App



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Write short notes on (any two)

a. LINQ

b. Open Redirected Attack

c. Query String

Expert approved answer

a) LINQ

LINQ (Language-Integrated Query) is a feature in .NET that allows developers to query and manipulate data using a unified syntax within the programming language. It provides a consistent way to work with data from various sources, such as databases, XML, collections, and more.

Here's a simple example of LINQ using LINQ to Objects in .NET:

using System;

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public class Program





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```
using System;
using System.Linq;
public class Program
{
    public static void Main()
    {
        int[] numbers = { 1, 2, 3, 4, 5 };
        // LINQ query expression
        var evenNumbersQuery = from num in
            numbers where num % 2 == 0 select num;
        var evenNumbers =
            evenNumbersQuery.ToList();
        // Display the results
        Console.WriteLine("Even Numbers:");
        foreach (var num in evenNumbers)
        {
            Console.WriteLine(num);
        }
    }
}
```

b) Open Redirected Attacks

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b) Open Redirected Attacks

Open redirect attacks are a type of security vulnerability that occurs when a web application allows an attacker to redirect users to an arbitrary URL. This can be exploited by an attacker to trick users into visiting malicious websites or phishing pages, potentially leading to various types of attacks, such as stealing sensitive information, distributing malware, or gaining unauthorized access to user accounts. The vulnerability typically arises when a web application includes a parameter in the URL that determines the destination of a redirect. If this parameter can be manipulated by an attacker, they can change the redirect URL to an external site under their control.

c) Query String

In .NET, a query string refers to the portion of a URL that contains data in the form of key-value pairs. It is typically used to pass

information from the client to the server.

b)



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appended to the URL and starts with a



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a URL that contains data in the form of key-value pairs. It is typically used to pass information from a client (such as a web browser) to a server. The query string is appended to the URL and starts with a question mark “?” followed by one or more key-value pairs separated by ampersands “&”.

Here's a simple example in .NET to illustrate the usage of a query string:

```
using System;
class Program
{
    static void Main()
    {
        string url = "http://example.com/page?
        param1=value1&param2=value2";
        // Parse the query string
        var query = new System.Uri(url).Query;
        var queryParams =
        System.Web.HttpUtility.ParseQueryString(query);
        // Retrieve values from the query string
```

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// Retrieve the values



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```
static void Main()
{
    string url = "http://example.com/page?
    param1=value1&param2=value2";
    // Parse the query string
    var query = new System.Uri(url).Query;
    var queryParams =
    System.Web.HttpUtility.ParseQueryString(query);
    // Retrieve values from the query string
    string param1 = queryParams["param1"];
    string param2 = queryParams["param2"];
    // Display the values
    Console.WriteLine("param1: " + param1);
    Console.WriteLine("param2: " + param2);
}
}
```

In this example, we start with a sample URL that contains a query string:

```
`http://example.com/page?
param1=value1&param2=value2`.
```



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